

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0497

Vc Porterdale Hydroelectric 138 kV
August 19, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0497. The Interconnection Customer proposes a 232.0 MW battery energy storage facility interconnecting at the Vc Porterdale Hydroelectric 138 kV substation in the Virginia area. The request was received on July 06, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 274 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 40 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$75,752,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lone Tree / Craven County Wood Energy Lp 138kV	Bear / Clover 345kV	168.7	226	14.2
Kerr Dam / Tecumseh Energy Ctr 345kV	North Tower Tap / McCartney 345kV	164.5	1,116	6.3
Craven County Wood Energy Lp Substation / Lowmoor 138kV L	Bear Substation / Clover 345kV	163.7	230	14.2
Franklin-Elk Grove Substation / New London 138kV	Monel / Candelaria Substation 138kV	140.1	222	8.8
Bear / Mecklenburg Power Station 345kV	Bear / Clover 345kV L	127.5	1,927	20.5
Vc Porterdale Hydroelectric / Bonsack 138kV	Craven County Wood Energy Lp / Lone Tree 138kV	125.0	187	54.8
Muddy Ranch / Roanoke Electric Steel 138kV L	Huntington Court 345 kV/138 kV Autotransformer	122.8	239	50.7
Tecumseh Energy Ctr / Gaston 345kV line	North Tower Tap / McCartney 345kV	122.6	1,352	7.2
Vc Porterdale Hydroelectric / Neely 138kV	Huntington Court / Reuses 345kV	119.1	202	42.1
Sharyland / Bixby Hill 345kV	Tombstone / Bixby Hill 500kV	118.9	2,557	6.0
Tombstone / Sharyland Substation 345kV	Tombstone / Bixby Hill 500kV Line	118.3	2,612	5.9
Blue Ride / Neely Substation 138kV	Reusens / Huntington Court Substation 345kV	118.1	217	42.1
Michigan Power Lp / Blue Ridge 138kV	Lone Tree / Craven County Wood Energy Lp 138kV line	114.0	322	20.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N Island / Clearbrook 138kV Line	Reusens / Leesville 345kV	113.7	302	14.4
North Island Substation / Shutesbury 138kV	Reusens / Leesville 345kV	112.1	312	14.4
Huntington Court / Bonsack 138kV line	Michigan Power Lp / Blue Ridge 138kV	112.1	219	65.8
Monel / Rustburg 138kV L	Clover / Bear 345kV	111.9	421	5.5
Safeway Stores / Shutesbury 138kV	Reusens / Leesville 345kV line	111.2	326	14.4
S Creek / Aibonito Pds 138kV	Bear / Clover 345kV line	110.0	372	5.8
Brush Tavern / Rustburg 138kV	Bear / Clover 345kV	109.8	460	5.5
Rally Hill / Train Substation 138kV line	Reusens / Leesville 345kV	109.2	250	10.9
Sierra Pacific Lumber / Riva Rad 345kV	Taber / Glenannan 345kV	108.1	1,495	5.5
Sierra Pacific Lumber 345/1kV Transformer Bank	Taber / Glenannan 345kV	108.1	1,495	5.5
Sierra Pacific Lumber 138 kV/1 kV Auto	Glenannan / Taber 345kV	108.1	1,495	5.5
Tiger Bay / Dranesville 345kV line	Taber / Glenannan 345kV Line	107.5	1,183	9.3
Jack Ramey Wind / Clearbrook 138kV	Reusens / Leesville 345kV	107.4	296	14.4
Rally Hill / Meadowview 138kV line	Reusens Substation / Leesville 345kV	106.3	246	10.9
Deset Outpost / South Texa 138kV	Brush Tavern / Rustburg 138kV	105.9	413	7.9
Train / Asheville 138kV	Reusens / Leesville 345kV Line	105.6	273	10.9
Fredericksburg / Possum Point 345kV	Nauvoo / Birchwood 345kV Line	104.9	1,326	6.1
Bear / Clover 345kV line	Bear / Mecklenburg Power Sta 345kV	103.4	2,416	26.9
Louisa Generation Facility Substation / Fluvanna P.S Substation 345kV	Leavitt / Birchwood 345kV	102.7	2,485	16.5
Hancock / Starkey 138kV line	Reuses / Leesville 345kV L	102.5	263	14.4
Bear / Leavitt 345kV L	Leavitt Substation / Bear 345kV #2	102.3	1,903	15.3
Glenannan / Taber 345kV Line	Sierra Pacific Lumber 138/1kV Transformer Bank	102.3	3,447	15.5
Hancock / Roanoke Electric Steel 138kV	Huntington Court 345 kV/138 kV TX	102.1	317	50.7
Sunbright / Fairland 138kV	Taber / Glenannan 345kV L	101.0	1,350	5.8
Starkey / Jack Ramey Wind 138kV	Leesville / Reusens 345kV	100.9	295	14.4
Sunbright / Colesville 138kV L	Taber / Glenannan 345kV	100.4	1,413	5.8
Poages Mill / Meadowview 138kV	Reusens / Leesville 345kV Line	100.2	232	10.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lone Tree / Craven County Wood Energy Lp 138kV line	Rebuild line with higher-capacity conductor	\$12,488,000	23% (\$2,862,000)
Kerr Dam Substation / Tecumseh Energy Ctr 345kV	Rebuild line with higher-capacity conductor	\$53,310,000	6% (\$3,322,000)
Craven County Wood Energy Lp Substation / Lowmoor 138kV L	Rebuild line with higher-capacity conductor	\$14,116,000	23% (\$3,235,000)
Franklin-Elk Grove / New London 138kV L	Rebuild line with higher-capacity conductor	\$2,946,000	8% (\$238,000)
Bear / Mecklenburg Power Sta 345kV line	Rebuild line with higher-capacity conductor	\$183,064,000	6% (\$11,806,000)
Vc Portedale Hydroelectric / Bonsack 138kV	Rebuild line with higher-capacity conductor	\$4,001,000	100% (\$4,001,000)
Muddy Ranch / Roanoke Electric Steel 138kV L	Rebuild line with higher-capacity conductor	\$1,137,000	84% (\$961,000)
Tecumseh Energy Ctr / Gaston 345kV line	Rebuild line with higher-capacity conductor	\$60,005,000	6% (\$3,693,000)
Vc Porterdale Hydroelectric / Neely 138kV L	Rebuild line with higher-capacity conductor	\$2,834,000	77% (\$2,180,000)
Sharyland Substation / Bixby Hill 345kV	Rebuild line with higher-capacity conductor	\$74,149,000	3% (\$2,432,000)
Tombstone / Sharyland Substation 345kV	Rebuild line with higher-capacity conductor	\$82,141,000	3% (\$2,708,000)
Neely / Blue Ridge 138kV L	Rebuild line with higher-capacity conductor	\$5,945,000	77% (\$4,573,000)
Michigan Power Lp / Blue Ridge 138kV	Rebuild line with higher-capacity conductor	\$9,273,000	100% (\$9,273,000)
N Island / Clearbrook 138kV Line	Rebuild line with higher-capacity conductor	\$13,116,000	35% (\$4,619,000)
North Island Substation / Shutesbury 138kV	Rebuild line with higher-capacity conductor	\$4,267,000	35% (\$1,503,000)
Huntington Court / Bonsack 138kV	Rebuild line with higher-capacity conductor	\$2,061,000	100% (\$2,061,000)
Monel / Rustburg 138kV Line	Rebuild line with higher-capacity conductor	\$12,035,000	4% (\$529,000)
Safeway Stores / Shutesbury 138kV	Rebuild line with higher-capacity conductor	\$4,974,000	35% (\$1,751,000)
South Creek / Aibonito Pds 138kV Line	Reconductor line	\$2,363,000	7% (\$174,000)
Brush Tavern / Rustburg 138kV	Reconductor line	\$2,347,000	4% (\$103,000)
Rally Hill / Train 138kV Line	Reconductor line	\$3,630,000	39% (\$1,420,000)
Sierra Pacific Lumber / Riva Rad 345kV	Reconductor line	\$12,786,000	3% (\$414,000)
Sierra Pacific Lumber 345 kV/1 kV TX	Replace transformer with higher-rated unit	\$16,625,000	3% (\$538,000)
Sierra Pacific Lumber 138 kV/1 kV Auto	Replace transformer with higher-rated unit	\$3,138,000	3% (\$102,000)
Tiger Bay / Dranesville 345kV line	Reconductor line	\$10,295,000	3% (\$303,000)
Jack Ramey Wind / Clearbrook 138kV	Reconductor line	\$795,000	35% (\$280,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Meadowview / Rally Hill Substation 138kV Line	Reconductor line	\$1,608,000	39% (\$629,000)
Desert Outpost / South Texa 138kV	Reconductor line	\$656,000	7% (\$48,000)
Train / Asheville 138kV	Reconductor line	\$5,037,000	39% (\$1,971,000)
Fredericksburg / Possum Point 345kV	Reconductor line	\$15,095,000	3% (\$505,000)
Bear / Clover 345kV line	Reconductor line	\$29,767,000	7% (\$2,016,000)
Fluvanna P.S Substation / Louisa Generation Facility 345kV	Reconductor line	\$26,509,000	4% (\$1,112,000)
Hancock / Starkey 138kV	Reconductor line	\$1,255,000	35% (\$442,000)
Bear / Leavitt 345kV L	Reconductor line	\$42,650,000	5% (\$2,261,000)
Taber / Glenannan 345kV	Reconductor line	\$15,039,000	3% (\$470,000)
Hancock / Roanoke Electric Steel 138kV	Reconductor line	\$527,000	84% (\$445,000)
Sunbright / Fairland 138kV	Reconductor line	\$159,000	3% (\$5,000)
Jack Ramey Wind / Starkey Substation 138kV	Reconductor line	\$1,006,000	35% (\$354,000)
Sunbright / Colesville 138kV	Reconductor line	\$442,000	3% (\$14,000)
Meadowview Substation / Poages Mill 138kV	Reconductor line	\$1,021,000	39% (\$399,000)
	Total allocated to project		\$75,752,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.